

Protothreads

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- Protothreads are extremely lightweight stackless threads designed for severely memory constrained systems, such as small embedded systems or wireless sensor network nodes.
- Protothreads provide linear code execution for event-driven systems implemented in C.
- Protothreads can be used with or without an underlying operating system to provide blocking event-handlers.
- Protothreads provide sequential flow of control without complex state machines or full multi-threading.

Example pthreads code

```
#include "pt.h"
struct pt pt;
struct timer timer;

PT_THREAD(example(struct pt *pt))
{
    PT_BEGIN(pt);

    while(1) {
        if(initiate_io()) {
            timer_start(&timer);
            PT_WAIT_UNTIL(pt, io_completed() || timer_expired(&timer));
            read_data();
        }
    }
    PT_END(pt);
}
```

- While protothreads originally were created for memory-constrained embedded systems, it has found many uses as a general purpose library too.
- Examples include multimedia streaming server software, grid computing research software, and MPEG decoding software for Internet TVs.

Main features:

- Very small RAM overhead - only two bytes per protothread and no extra stacks
- Highly portable - the protothreads library is 100% pure C and no architecture specific assembly code
- Can be used with or without an OS
- Provides blocking wait without full multi-threading or stack-switching
- Freely available under a BSD-like open source license

Example applications:

- Memory constrained systems
- Event-driven protocol stacks
- Small embedded systems
- Sensor network nodes
- Portable C applications